

1.4
(1-10, 14, 16) Find the general solution of the following equation.

1. $xy' = x + y$

$$\frac{dy}{dx} = \frac{x+y}{x} \Rightarrow \text{Homog. diff. eqn}$$

Put $y = vx$

$$\frac{dy}{dx} = v + xv'$$

$$\therefore v + x \frac{dv}{dx} = \frac{x+vx}{x} = 1+v$$

$$x \frac{dv}{dx} = 1$$

$$\Rightarrow \int dv = \int \frac{dx}{x}$$

$$v = \ln x + c$$

$$\Rightarrow \frac{y}{x} = \ln x + c$$

2) $x \frac{dy}{dx} = 2(x+y)$

$$\Rightarrow \frac{dy}{dx} = \frac{2(x+y)}{x} \Rightarrow \text{Homog.}$$

Put $y = vx \Rightarrow y' = v + xv'$

$$\therefore v + xv' = \frac{2(x+vx)}{x} = \frac{2x(1+v)}{x}$$

$$v + x \frac{dv}{dx} = 2(1+v)$$

$$x \frac{dv}{dx} = 2 + 2v - v$$

$$x \frac{dv}{dx} = v + 2$$

$$\Rightarrow \int \frac{dx}{x} = \int \frac{dv}{v+2}$$

$$\ln x = \ln(v+2) + \ln c$$

$$x = c(v+2)$$

$$x = c\left(\frac{y}{x} + 2\right)$$

$$x^2 = c(y+2x)$$

3) $xyy' = \frac{1}{2}(y^4 + x^4)$

$$\frac{dy}{dx} = \frac{(y^4 + x^4)}{2xy} \Rightarrow \text{Homog.}$$

Put $y = vx \Rightarrow y' = v + xv'$

$$\therefore v + xv' = \frac{v^4 x^4 + x^4}{2vx^2}$$

$$v + xv' = \frac{x^2(1+v^4)}{2vx^2}$$

$$v + xv' = \frac{v^4 + 1}{2v}$$

$$x \frac{dv}{dx} = \frac{v^2 + 1}{2v} - v$$

$$x \frac{dv}{dx} = \frac{v^2 + 1 - 2v^2}{2v}$$

$$x \frac{dv}{dx} = \frac{1 - v^2}{2v} = \frac{v^2 - 1}{2v}$$

$$\frac{dx}{x} = \int \frac{2v}{v^2 - 1} dv \Rightarrow \text{V. Separable}$$

$$\ln x = \ln(v^2 - 1) + \ln c$$

$$\ln x = \ln\left(\frac{y^2 - x^2}{x^2}\right) + \ln c$$

$$\Rightarrow x = \frac{(y^2 - x^2)^c}{x^2}$$

$$\Rightarrow x^3 = c(y^2 - x^2)$$

$$) \quad x^2 y' = y^2 + 5xy + 4x^2$$

$$y' = \frac{y^2 + 5xy + 4x^2}{x^2} \Rightarrow \text{Homog.}$$

$$+xv' = \frac{v^2 x^2 + 5vx^2 + 4x^2}{x^2}$$

$$+xv' = v^2 + 5v + 4$$

$$xv' = v^2 + 4v + 4$$

$$xv' = (v+2)^2$$

$$x \frac{dv}{dx} = (v+2)^2$$

$$\Rightarrow \frac{dx}{x} = \frac{dv}{(v+2)^2}$$

$$\Rightarrow \frac{dx}{x} = (v+2)^{-2} dv$$

$$\ln x = \frac{(v+2)^{-1}}{-1-1} + c$$

$$\ln x = -\frac{1}{v+2} + c$$

$$\ln x = -\frac{x}{y+2x} + c$$

$$(5) \quad x^2 y' = y^2 + xy + x^2$$

$$\frac{dy}{dx} = \frac{y^2 + xy + x^2}{x^2} \Rightarrow \text{Homog.}$$

$$v + x \frac{dv}{dx} = v^2 + v + 1$$

$$\Rightarrow x \frac{dv}{dx} = v^2 + 1$$

$$\Rightarrow \int \frac{dx}{x} = \int \frac{dv}{1+v^2}$$

$$\ln x + \ln c = \tan^{-1} v$$

$$\ln xc = \frac{\tan^{-1} y}{x}$$

$$(6) \quad (xy' - y) \cos\left(\frac{y}{x}\right) = -3x^4$$

$$\text{Put } y = xv \Rightarrow y' = v + xv'$$